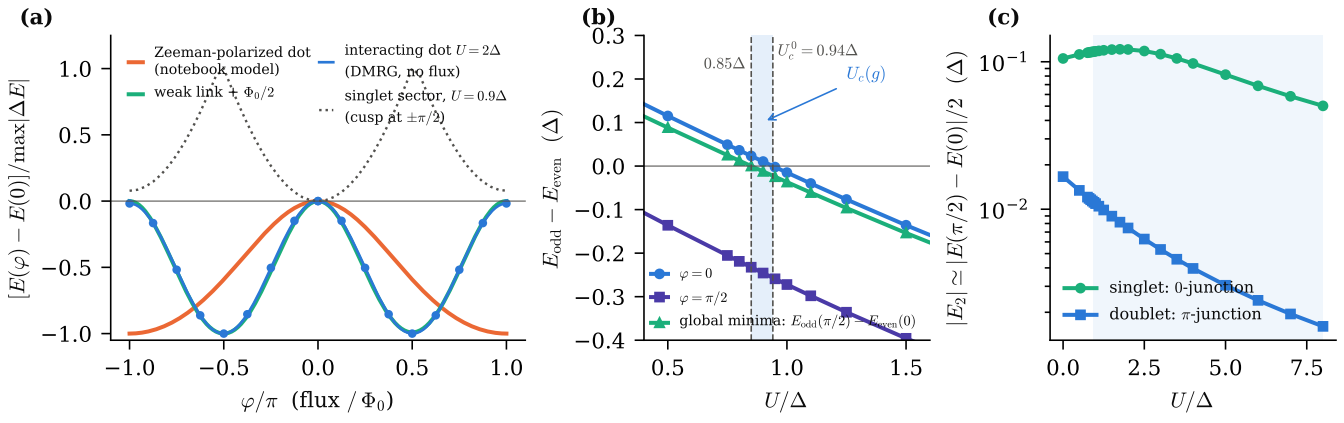
(a)
Zeeman-polarized dot
(notebook model)
weak link + $\Phi_0/2$
interacting dot $U = 2\Delta$
(DMRG, no flux)
singlet sector, $U = 0.9\Delta$
(cusp at $\pm\pi/2$)
$[E(\varphi) - E(0)]/\max|\Delta E|$
φ/π (flux / Φ_0)
(b)
0.85Δ
$U_c^0 = 0.94\Delta$
$U_c(g)$
$\varphi = 0$
$\varphi = \pi/2$
global minima: $E_{\rm odd}(\pi/2) - E_{\rm even}(0)$
$E_{\rm odd} - E_{\rm even}$ (Δ)
U/Δ
(c)
$|E_2| \simeq |E(\pi/2) - E(0)|/2$ (Δ)
singlet: 0-junction
doublet: π-junction
U/Δ